Chinmaya B J

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Profile

Developer Consultant at SAP India building production AI agents and LLM-powered automation on SAP BTP. B.E. graduate in Artificial Intelligence and Machine Learning with a published research record in cybersecurity and applied AI. Eager to keep learning emerging technologies and deepen my technical expertise to build scalable, impactful solutions.

Experience

SAP Developer Consultant — Bengaluru

SAP India (Business Data Cloud) — *Jul 2026 – Present*

- Built an LLM-powered platform (Flask, React, `@sap/datasphere-cli`) that deploys SAP Datasphere master-data objects (GL Account, Cost Center, Profit Center) with up to 27 artifacts per run, replacing a manual, expert-gated 3–4 hour process.
- Added Claude-based CSN enrichment (∼70% less manual annotation) and a voice + text agentic copilot that executes wizard actions and deployments, cutting time-to-first-deploy from ∼3.5 hours to under 5 minutes.
- Automated S/4HANA → Datasphere Replication Flow setup in a single CLI call, with a BTP Audit Log trail for GDPR/SOX traceability and no credentials reaching model context.
- Converted a manual SharePoint-based agent tracking workflow into an automated AI Agent Catalog on SAP BTP Cloud Foundry (Flask, React, PostgreSQL, MS Graph) with semantic search and an approval workflow; a 15-minute automated sync cut data staleness from 2–3 days to under 30 minutes, saving ∼5 hours/week of manual upkeep across 60+ governed agents.

Samsung Research Intern — Bangalore

Samsung R&D — *Jul 2025 – Jan 2026*

- Engineered a lightweight animation pipeline using Stable Diffusion v1.5 with DreamBooth LoRA for stylized video synthesis.
- Optimised inference throughput from 12 fps to 18 fps on mid-range mobile SoCs via quantisation and batching.
- Benchmarked deployment on mobile devices and platforms like Replicate, OpenXLab.

Dell Project Intern — Bangalore

Dell Technologies — *Jul 2024 – May 2025*

- Simulated 20+ ransomware variants (Lockbit, Ryuk, WannaCry) to benchmark detection; measured TTD, accuracy and encryption speed.
- **Collaborated directly with Dell's Senior Architect** to align simulator design with enterprise threat-modelling framework and secure sign-off for production hardening.

Outspark Intern — Remote

Outspark — *Jul 2023 – Dec 2023*

- Developed a voice-enabled resume builder using spaCy, NLTK, BERT, and speech recognition for dynamic form-filling.
- Extracted structured data (name, education, experience) from natural-language resumes; delivered a working Proof of Concept for future productisation.

Projects

Single-Lead ECG Arrhythmia Classification (CAT-Net) | *Python, TensorFlow/PyTorch, CNN, Attention, Transformer, MIT-BIH,*

- Authored an end-to-end ML pipeline on the MIT-BIH Arrhythmia Database using single-lead (MLII) ECG: wavelet (DWT) denoising, R-peak detection, heartbeat segmentation, and train/validation/test splitting (80/20, with 15% of training held out for validation).
- Addressed severe class imbalance by benchmarking SMOTEENN, ADASYN and SMOTETomek, and selected SMOTETomek for training-set balancing.
- Designed and trained CAT-Net, a custom CNN + Attention + Transformer model of 1.8M parameters, and compared it against standalone Transformer and CNN-LSTM baselines.
- Tuned filter count, transformer encoder layers, attention heads, and transformer/FFN dimensions, reducing validation loss and improving validation/test accuracy, F1 and per-class scores.
- Classified 5 arrhythmia classes (N, S, V, F, Q) with 97% average accuracy and 93% average F1 score across all classes.

Hydrogen (H_2) Leak Detection & Anomaly Analysis | *Academic Research* | *Python, scikit-learn, TensorFlow/Keras (LSTM), Panda*

- Built an end-to-end ML pipeline on a time-synchronized 1 Hz multi-sensor dataset (405 samples, 24 channels: H_2 concentration, pressure, mass flow rate, humidity) from an EU hydrogen-safety research experiment, for early leak detection and predictive maintenance.
- Authored the feature-engineering module: 13 physics-informed features (first/second-order derivatives, rolling-window and trend statistics, pressure–flow and pressure–temperature ratios, deviation from baseline); built leak and severity (warning/critical) labels from H_2 thresholds.
- Diagnosed that every sample exceeded the 1% leak threshold (single-class data), making binary classification invalid, and reframed the task as H_2 concentration regression plus unsupervised anomaly detection.

- Showed through multivariate drift analysis (distance from a data-driven normal baseline, lowest 20% of H_2 readings) that leak conditions appear as multi-sensor patterns rather than single-sensor threshold breaches.
- Trained Random Forest (5-fold CV $R^2 = 0.996$, RMSE 0.55 on a 1.59–34.5% range) and an LSTM sequence regressor; random-search tuning (seq. length 10, 64 units) cut LSTM CV RMSE by ∼48% (1.03 → 0.53; holdout RMSE 0.54).
- Applied Isolation Forest (10% contamination) to flag 33/324 training and 15/81 test samples as anomalies without ground-truth labels, and exported the model and scaler for reuse.
- Ranked features with five importance methods (impurity, permutation, drop-column, mutual information, Spearman) plus LSTM permutation/occlusion; CH3 pressure was the strongest predictor, followed by CH2 mass flow rate and humidity.

Real-Time Audio Fingerprinting for Sung Mantras | *Audio Signal Processing, Audio Fingerprinting, Real-Time Systems*

- Co-developed a real-time system that identifies and verifies sung mantras in audio recordings by converting signals into compact, discriminative "fingerprints" stored in a database and matched against incoming audio streams.
- Designed the fingerprinting approach to stay accurate despite variations in pitch, speed and vocal characteristics, and to tolerate noise, minor distortion and overlapping sounds, where traditional audio-matching methods struggle.
- Targeted use cases include meditation and spiritual practice, content recognition on multimedia platforms, and voice-based indexing of mantra repositories; work carried out with a team at R.V. College of Engineering under Dr. Lingayya Hiremath and published as a research paper.

BrightSide: AI Debate Coaching with Emotional Intelligence | *BERT, GPT, NLP, Sentiment Analysis, Web*

- Built a real-time, web-based platform that pairs end-to-end debate coaching with emotional-intelligence training, addressing the gap in conventional education that focuses on articulation and logic but overlooks emotional communication.
- Integrated sentiment analysis, NLP, and a transformer-based chatbot (GPT, BERT) to assess argument structure, emotional tone, and empathy in user interactions.
- Introduced novel feedback metrics such as the user's voice pitch, delivering real-time feedback; simulations and experiments showed improved emotional control and higher student engagement.
- Published as "BrightSide: Real-Time Emotional Intelligence Meets Debate Coaching" (Nov 2025, IEEE).

Paper Publications

- **BrightSide: Real-Time Emotional Intelligence Meets Debate Coaching** (Nov 2025) — AI debate-coaching platform combining BERT/GPT-based sentiment analysis and voice-pitch metrics to train emotional intelligence. View Publication
- **Real Time Audio Fingerprinting System for Identifying Sung Mantras** (with K. Patil, A. A. Dharmavaram, A. M. Athreya, Dr. L. Hiremath; R.V. College of Engineering) — noise-tolerant, real-time audio fingerprinting for identifying and verifying sung mantras despite variations in pitch, speed and vocal characteristics.
- **Targeted Ransomware Attacks and Detection to Strengthen Cybersecurity Strategies** (IEEE ICACRS 2023) — cited by 7 papers across international journals and conferences, including *Forensic Science International: Digital Investigation.* View Publication

Technical Skills

Languages: Python, C, HTML, CSS, JavaScript, Bash

Frameworks & Tools: Flask, Django, DRF, Streamlit, React, TensorFlow, Keras, PyTorch, scikit-learn, OpenCV, Stable Diffusion, DreamBooth, HuggingFace, NLP, spaCy, NLTK, BERT, Google Colab

ML & Data: CNN, LSTM, Attention/Transformers, Time-Series Feature Engineering, Anomaly Detection (Isolation Forest), Random Forest, Signal Processing (DWT), Audio Fingerprinting, Imbalanced Learning (SMOTETomek, SMOTEENN, ADASYN), Hyperparameter Tuning, Cross-Validation, Model Explainability, Sentiment Analysis, Speech Recognition, Pandas, NumPy, Matplotlib, Seaborn

Other: SQL, PostgreSQL, Linux, GenAI, LLM APIs (Claude, Gemini), SAP BTP, SAP Business Data Cloud (BDC), SAP Datasphere, Cryptography, Git, REST APIs

Certifications

- **Coursera:** Advanced Learning Algorithms; Supervised Machine Learning (100% grade)

Education

R.V. College of Engineering — Bengaluru
B.E. in AI & ML — CGPA: 9.18 — *2022 – 2026*

Sri Chaitanya PU College — Bengaluru
CBSE (Class XII) — 91% — *2021 – 2022*

Sri Chaitanya School — Bengaluru
CBSE (Class X) — 95% — *2019 – 2020*